HOSPITAL MANAGEMENT SYSTEM

A Patient Record and Appointment Management System

Assumptions

The Hospital Management System is assumed to be used by a hospital to manage its general activities efficiently. The hospital consists of multiple departments, doctors, nurses, staff, and patients. Patients can register with the hospital, consult doctors, and schedule appointments based on doctor availability. The system is assumed to maintain necessary information related to patients, doctors, departments, appointments, treatments, and medical records.

The system is also assumed to handle hospital billing and payment activities. Bills may be generated for consultations, tests, treatments, medicines, and other services, and payments can be recorded accordingly. Different users such as administrators, doctors, receptionists, and patients may have different responsibilities and access to the system. The system is expected to provide accurate, secure, reliable, and easily accessible information to support the day-to-day functioning of the hospital.

Functional Requirements

ID	Requirement	Description
FR-1	User Authentication & Role Management	The system should provide login for Admin, Doctor, Receptionist, and Patient using valid credentials and display a role-specific dashboard.
FR-2	Patient & Medical Record Management	The system should register patients and maintain their personal details, medical history, diagnosis, treatment, and prescriptions.
FR-3	Doctor & Department Management	The system should manage doctor details, specializations, availability, and departments, including assigning doctors to departments.
FR-4	Appointment Management	The system should allow patients or staff to book, reschedule, and cancel appointments based on doctor availability.
FR-5	Admission & Discharge Management	The system should manage patient admission, room/bed allocation, treatment during hospitalization, and discharge details.
FR-6	Billing & Payment Management	The system should generate bills for consultations, tests, treatments, medicines, and other services and record the corresponding payments.
FR-7	Notification & Reminder Management	The system should provide notifications or reminders for appointments, scheduled tests, payments, and other important hospital activities.
FR-8	Report Management	The system should generate reports related to patients, doctors, appointments, medical records, admissions, laboratory tests, and payments for hospital management.

Non-Functional Requirements

ID	Requirement	Description
NFR-1	Security	The system should protect patient, medical, and payment information from unauthorized access using authentication and role-based access control.
NFR-2	Performance	The system should respond quickly to operations such as patient searches, appointment booking, record retrieval, and bill generation.

ID	Requirement	Description
NFR-3	Reliability	The system should operate consistently without frequent failures and should ensure that important hospital data is not lost or corrupted.
NFR-4	Availability	The system should be accessible whenever required by hospital staff and should minimize downtime during hospital operations.
NFR-5	Usability	The system should provide a simple, clear, and user-friendly interface that allows hospital staff to perform tasks easily with minimal training.
NFR-6	Scalability	The system should be capable of supporting an increasing number of patients, doctors, departments, appointments, and transactions without major performance issues.
NFR-7	Maintainability	The system should be easy to maintain, modify, debug, and update when new requirements or features are introduced.
NFR-8	Data Integrity	The system should maintain accurate, consistent, and complete patient, medical, appointment, and payment information.
NFR-9	Backup and Recovery	The system should regularly back up important data and provide mechanisms to recover information in case of system failure or data loss.
NFR-10	Compatibility	The system should function correctly across commonly used devices, operating systems, browsers, and other hospital software when required.